

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

EDU-12: Student Query Intent Classification

1. Introduction

1.1 Purpose

The purpose of this project is to develop an intelligent educational query classification system that understands the learning intent behind student questions and categorizes them into predefined educational intents.

The system helps educational platforms understand what type of academic assistance a student requires, enabling better organization and analysis of student learning needs.

1.2 Project Title

EduIntent: Intelligent Student Query Intent Classification System

1.3 Scope

EduIntent is an AI-powered educational query analysis system that:

- Classifies student queries based on educational intent.
 - Identifies the associated subject and topic.
 - Detects multiple intents within a query.
 - Maintains query history.
 - Provides learning analytics.
 - Supports future expansion into intelligent educational systems.
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2. Problem Statement

Students ask academic questions in different ways depending on their learning requirements.

Examples:

- “What is polymorphism?” → Definition
- “Explain polymorphism.” → Explanation

- “Give an example of polymorphism.” → Example
- “Compare TCP and UDP.” → Comparison
- “Solve this equation.” → Problem Solving

Most educational systems process these queries similarly without understanding the student’s actual requirement.

As a result:

- Students may not receive appropriately categorized assistance.
- Educational platforms cannot effectively analyze learning needs.
- Valuable insights about student learning behavior are lost.

The proposed system solves this problem by automatically identifying the educational intent, subject, and topic of a student’s query.

3. Objectives

The objectives of the system are:

- Automatically identify educational intent.
- Classify student queries into predefined categories.
- Detect multiple intents in a single query.
- Identify subject and topic information.
- Store query history.
- Generate educational analytics.
- Improve educational query understanding.
- Support future personalized learning systems.

4. Functional Requirements

FR1 – Query Submission

The system shall allow students to submit academic questions through a web-based interface.

Example

Input:

“What is polymorphism?”

Output:

“Query submitted successfully.”

FR2 – Query Preprocessing

The system shall preprocess submitted queries before classification.

The preprocessing process shall include:

- Text cleaning
 - Tokenization
 - Stop-word removal
 - Normalization
 - Feature extraction
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FR3 – Intent Classification

The system shall classify student queries into one or more of the following educational intents:

Intent ID	Intent
INT-01	Definition
INT-02	Explanation
INT-03	Example
INT-04	Problem Solving
INT-05	Revision
INT-06	Comparison

Example

Input:

“What is polymorphism?”

Output:

Intent: Definition

FR4 – Subject Identification

The system shall identify the academic subject associated with the query.

Example

Input:

“What is the difference between TCP and UDP?”

Output:

Subject: Computer Networks

FR5 – Topic Detection

The system shall identify the specific topic discussed in the query.

Example

Output:

Subject: Computer Networks

Topic: TCP and UDP

FR6 – Multi-Intent Detection

The system shall identify multiple educational intents when a query contains more than one learning requirement.

Example

Input:

“Explain normalization, compare 2NF and 3NF, and give an example.”

Output:

- Explanation
 - Comparison
 - Example
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FR7 – Query History Management

The system shall store previous queries and classification results.

Stored information includes:

- Query Text
 - Intent
 - Subject
 - Topic
 - Timestamp
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FR8 – Analytics Dashboard

The system shall provide analytics including:

- Total number of queries processed
 - Intent distribution
 - Subject-wise query count
 - Frequently searched topics
 - Most requested educational intent
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5. Non-Functional Requirements

5.1 Performance

- Query classification shall complete within 2–3 seconds.
- The system shall support multiple concurrent users.

5.2 Usability

- The interface shall be simple and student-friendly.
- Results shall be displayed clearly and accurately.

5.3 Reliability

- Similar queries shall produce consistent classifications.
- The system shall operate reliably during demonstrations.

5.4 Scalability

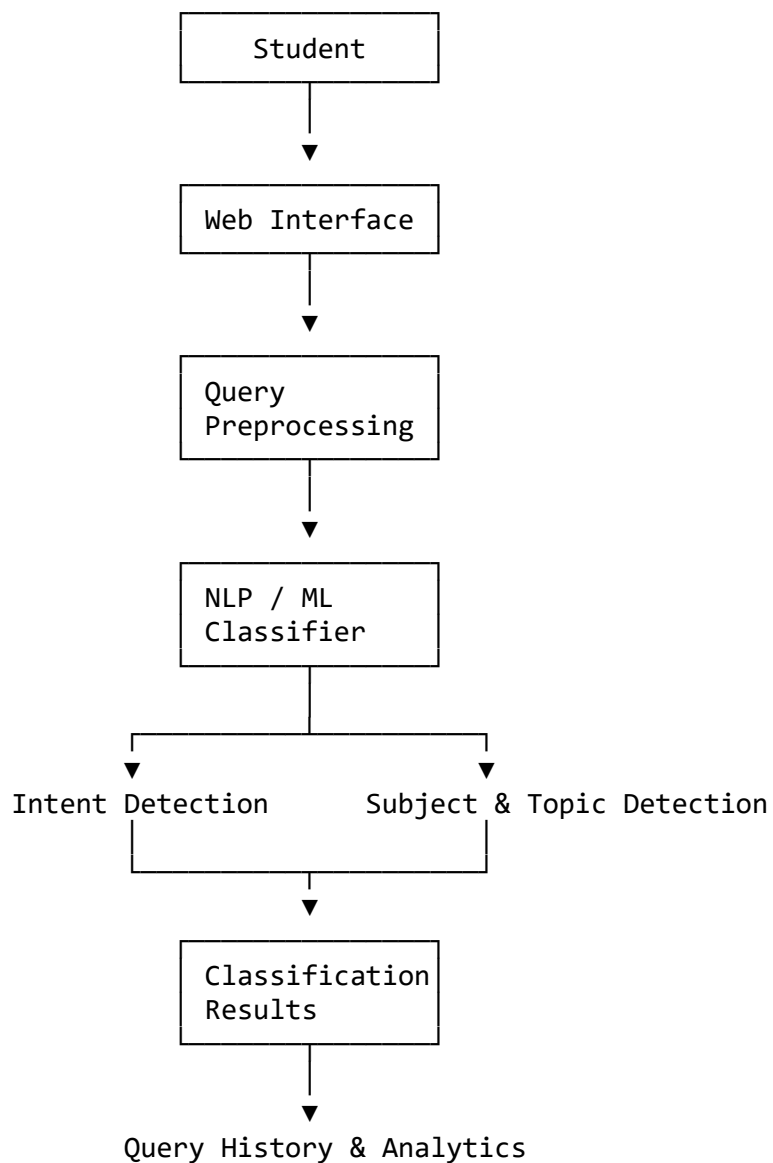
- Additional educational intents
- Additional academic subjects

- Larger datasets
- Future AI integrations

5.5 Security

- User data shall be protected.
 - Query history shall be stored securely.
 - Authentication may be implemented for registered users.
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6. System Architecture



7. Technology Stack

Layer	Technology
Frontend	React.js
Backend	FastAPI (Python)
NLP	NLTK, spaCy
Machine Learning	Scikit-learn
Feature Extraction	TF-IDF
Classification Model	Logistic Regression / SVM
Database	MongoDB / PostgreSQL
Deployment	Vercel + Railway/Render

8. Dataset Requirements

The dataset shall contain academic queries labeled according to educational intent.

Query	Intent
What is polymorphism?	Definition
Explain polymorphism.	Explanation
Give an example of polymorphism.	Example
Solve this equation.	Problem Solving
Give revision notes for DBMS.	Revision
Compare TCP and UDP.	Comparison

Subjects Covered

- Computer Networks
 - DBMS
 - Data Structures
 - Operating Systems
 - Java
 - Python
 - Mathematics
 - Machine Learning
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9. Machine Learning Workflow

Labeled Dataset



Data Cleaning



Train-Test Split



TF-IDF Feature Extraction



Model Training



Intent Classification



Model Evaluation



Deployment

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

10. User Roles

Student

- Submit academic queries
- View classification results
- View query history

Admin / Teacher

- Monitor analytics
 - Review query statistics
 - Manage training data
 - Add new categories
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11. Key Innovation

Educational Intent Classification

The system identifies the actual learning requirement behind a student's question.

Multi-Intent Recognition

The system can classify multiple educational intents from a single query.

Subject-Aware Analysis

The system identifies both:

- What the student wants
- Which academic topic is involved

Educational Analytics

The system provides insights into student learning patterns and query behavior.

12. Example Use Case

Input

“Explain DBMS normalization and compare 2NF with 3NF.”

Output

Detected Intents:

- Explanation
- Comparison

Subject:

DBMS

Topic:

Normalization

13. Expected Outcome

The developed solution will:

Understand



Classify



Identify Subject



Identify Topic



Analyze

The system enables educational platforms to better understand student learning requirements through intelligent query classification.

14. Future Scope

- Voice-based query classification
 - Multilingual support
 - Personalized learning recommendations
 - Difficulty-level prediction
 - Study Mode and Exam Mode
 - Student learning analytics
 - Adaptive learning pathways
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15. Conclusion

EduIntent is an AI-powered educational query intent classification system designed to understand student learning requirements more effectively. By combining NLP, machine learning, subject detection, topic detection, and multi-intent recognition, the system provides a strong foundation for future intelligent educational platforms.